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United States Patent

12403267

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

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Devices and methods for installation and removal of a needle tip of a needle

Abstract

The present disclosure provides medical devices, particularly medical grade needles with removable tips, as well as methods of needle tip installation and removal, and devices therefor. Needle tip installation and removal devices may include needle tip installation and removal tools and kits, including hand held and manipulated tools that facilitate the installation and removal of needle tips, particularly on vascular access port needles. Such installation and removal devices may provide more expedient hook up of the access port for the therapy. Also, the installation and removal of the needle tips may be conducted in a more sterile manner.

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Appl. No.: 18/135636

Filed: April 17, 2023

Prior Publication Data

Document Identifier

Publication Date

US 20240082510 A1

Mar. 14, 2024

Related U.S. Application Data

continuation parent-doc US 16724632 20191223 US 11628261 child-doc US 18135636

continuation parent-doc US 15301498 US 10512734 20191224 WO PCT/US2015/024256

Publication Classification

Int. Cl.: A61M5/34 (20060101); A61M39/02 (20060101); A61M5/142 (20060101)

U.S. Cl.:

CPC A61M5/343 (20130101); A61M5/346 (20130101); A61M39/0208 (20130101);
A61M39/0247 (20130101); A61M2005/14284 (20130101)

Field of Classification Search

CPC: A61M (2005/14284); A61M (39/0208); A61M (39/0247); A61M (5/343); A61M (5/346)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This non-provisional patent application is a continuation of U.S. application Ser. No. 16/724,632, filed Dec. 23, 2019, now U.S. Pat. No. 11,628,261, which is a continuation of U.S. application Ser. No. 15/301,498, filed Oct. 3, 2016, now U.S. Pat. No. 10,512,734, which is a national stage (35 U.S.C. 371) application of PCT patent application no. PCT/US2015/024256, filed Apr. 3, 2015, which claims the benefit of U.S. provisional application No. 61/974,807, filed Apr. 3, 2014.

FIELD

(1) The present disclosure relates to medical devices and more particularly to medical grade needles with removable tips, methods of needle tip installation and removal, and devices therefor.

BACKGROUND

(2) Medical patients, such as oncology patients, hemodialysis patients and hematology patients, may be subject to frequent fluid infusion treatments and/or fluid extraction treatments. Fluid infusion treatments may deliver medicaments (e.g. pharmaceutical products; therapeutic drugs), bodily fluid (e.g. blood), nutrients, contrasting agents, dialysis fluid and other liquid compositions to the body, while fluid extraction treatments may remove fluids such as dialysis fluid, bodily fluid (e.g. blood as part of phlebotomy) and other liquid compositions from the body. The fluid infusion treatment and fluid extraction treatment may be part of a fluid exchange treatment, such as dialysis.

(3) Many fluid treatments involve the use of an indwelling catheter with transgresses out of the body. However, where the catheter exits the body, there may be an increased risk of infection. In order to reduce the likelihood of infection, certain medical application may be able to utilize a vascular access port implanted beneath the cutaneous tissue (skin) of the patient/body.

(4) An implanted vascular access port may include an access point, such as a septum. The septum may be formed of a self-healing silicone material. External access to the implanted vascular access port may be accomplished by inserting the needle through the patient's skin and through the septum of the implanted port. However, a clinician needs to properly target the access port and, as a result, multiple needle sticks may be required to properly locate and access the access port, which may add discomfort to the patient. The access port is coupled to an indwelling catheter, which is inserted into a vein, such as a jugular vein, subclavian vein or superior vena cava.

SUMMARY

(5) Access to the implanted vascular access port may be accomplished by extending a hollow needle, particularly with a removable closed tip, from within the access port through the skin. The removable closed tip may keep the needle lumen closed to the ingress of body fluid while the port is not in use. Once the needle has raised (penetrated) through the skin, the removable closed tip may be removed to gain access to the lumen thereof for the infusion of fluids into the patient and/or extraction of fluids from the patient as part of a medical diagnostic and/or therapeutic treatment. Such needle tips may be replaced with a sterile tip to reduce the likelihood of infection to the patient.

(6) The present disclosure provides medical devices, particularly medical grade needles with removable tips, as well as methods of needle tip installation and removal, and devices therefor. Needle tip installation and removal devices may include needle tip installation and removal tools and kits, including hand held and manipulated tools that facilitate the installation and removal of needle tips, particularly on vascular access port needles. Such installation and removal devices may provide more expedient hook up of the access port for therapy. Also, the installation and removal of the needle tips may be conducted in a more sterile manner.

(7) In at least one embodiment, a medical device is provided which comprises at least one needle tip having a pointed closed end; and at least one tool configured to install the at least one needle tip on a needle shaft, the tool comprising a tool body having a proximal end and a distal end, and a needle tip holder located at the distal end of the tool, the needle tip holder configured to hold the needle tip during an installation of the needle tip on the needle shaft. The at least one needle tip and the at least one tool may be components of a sterilized medical device kit.

(8) In at least one embodiment, the at least one needle tip may be securable to the needle shaft by a positive mechanical engagement connection, such as provided by a threaded connection. In such instance, the at least one needle tip may comprise a needle shaft receptacle to receive a portion of the needle shaft; the needle shaft receptacle may include needle shaft receptacle threads; the needle shaft may include needle shaft threads; and the at least one needle tip may be securable to the

needle shaft by threaded engagement of the needle shaft receptacle threads of the needle tip with the needle shaft threads of the needle shaft.

- (9) In at least one embodiment, the needle tip holder may comprise a needle tip receptacle configured to receive the needle tip, and the needle tip receptacle may comprise a mating portion configured to mate with a mating portion of the needle tip.
- (10) In at least one embodiment, the mating portion of the needle tip receptacle may be configured to mate with the mating portion of the needle tip such that, when the tool is rotated about a longitudinal axis of rotation, both the needle tip receptacle and the needle tip rotate about the longitudinal axis of rotation.
- (11) In at least one embodiment, the mating portion of the needle tip receptacle may have a polygonal shape; and the mating portion of the needle tip may have a polygonal shape which fits inside the polygonal shape of the needle tip receptacle.
- (12) In at least one embodiment, the needle tip receptacle of the needle tip holder may be configured to hold the needle tip during the installation of the needle tip on the needle shaft.
- (13) In at least one embodiment, the needle tip receptacle may be configured to hold the needle tip during the installation of the needle tip on the needle shaft with a positive mechanical engagement connection formed with the needle tip.
- (14) In at least one embodiment, the at least one needle tip may include needle tip threads; the needle tip receptacle may include needle tip receptacle threads; and the positive mechanical engagement may be formed by threaded engagement of the needle tip threads with the needle tip receptacle threads.
- (15) In at least one embodiment, the at least one needle tip may include at least one needle tip ridge and/or at least one needle tip groove; In at least one embodiment, the needle tip receptacle includes at least one needle tip receptacle ridge and/or at least one needle tip receptacle groove; and the positive mechanical engagement may be formed by positive mechanical engagement of the at least one needle tip ridge with the at least one needle tip receptacle groove and/or engagement of the at least one needle tip groove with the at least one needle tip receptacle ridge.
- (16) In at least one embodiment, the positive mechanical engagement may be formed by a portion of the at least one needle tip which overlaps a portion of the needle tip receptacle in a plane transverse to a longitudinal axis of the tool.
- (17) In at least one embodiment, the needle tip receptacle may be configured to hold the needle tip during the installation of the needle tip on the needle shaft with a friction fit connection formed with the needle tip.
- (18) In at least one embodiment, the needle tip receptacle may include an elastically deformable member; and the friction fit connection may be formed by elastic deformation of the elastically deformable member when the needle tip is in the needle tip receptacle.
- (19) In at least one embodiment, the needle tip receptacle may be configured to hold the needle tip during the installation of the needle tip on the needle shaft with a magnetic force applied to the needle tip.
- (20) In at least one embodiment, the needle tip receptacle may include a magnet; and the needle tip may comprise a magnetic material.
- (21) In at least one embodiment, the needle tip receptacle may be configured to hold the needle tip during the installation of the needle tip on the needle shaft with a vacuum force applied to the needle tip.
- (22) In at least one embodiment, the needle tip receptacle may be in fluid communication with a vacuum passage through which a vacuum is drawn with a vacuum generating device.
- (23) In at least one embodiment, a method of medical treatment may be provided, with the method comprising extending a needle located beneath skin of a host such that the needle extends through the skin of the host from beneath the skin of the host, wherein the needle is coupled to an access port implanted beneath the skin of the host and the needle comprises a first removable needle tip

secured to a needle shaft; removing the first removable needle tip from the needle shaft with a first needle tip tool to expose a lumen of the needle; administering medical treatment to the host via the lumen of the needle; installing the first removable needle tip or a second removable needle tip on the needle shaft with the first needle tip tool or a second needle tip tool; and retracting the needle beneath the skin of the host.

Description

FIGURES

- (1) The above-mentioned and other features of this disclosure, and the manner of attaining them, may become more apparent and better understood by reference to the following detailed description of embodiments described herein taken in conjunction with the accompanying drawings, wherein:
- (2) FIG. 1A is a cross-sectional view of a medical device comprising an access port and a catheter; with a needle of the access port including a removable needle tip in a concealed (retracted) position relative to access port body;
- (3) FIG. 1B is a cross-sectional view of the embodiment of the medical device of FIG. 1A; with the needle of the access port in an exposed (extended) position relative to access port body and the removable needle tip removed;
- (4) FIG. 2 is a perspective view of an embodiment of a needle tip installation and removal tool according to the present disclosure, with the needle of the access port in alignment therewith;
- (5) FIG. 3 is a close-up cross-sectional side view of a distal end portion of the needle tip installation and removal tool of FIG. 2 joined with the needle of FIG. 2;
- (6) FIG. 4 is a close-up perspective view of a distal end portion of the needle tip installation and removal tool of FIG. 2 and the needle of FIG. 2;
- (7) FIGS. 5A and 5B are cross sectional side views of one embodiment of a needle tip installation and removal tool, and needle, according to the present disclosure, with the needle tip secured with the needle in FIG. 5A, and the needle tip removed from the needle and held by the tool in FIG. 5B;
- (8) FIGS. 6A and 6B are cross sectional side views of another embodiment of a needle tip installation and removal tool, and needle, according to the present disclosure, with the needle tip secured with the needle in FIG. 6A, and the needle tip removed from the needle and held by the tool in FIG. 6B;
- (9) FIGS. 7A and 7B are cross sectional side views of another embodiment of a needle tip installation and removal tool, and needle, according to the present disclosure, with the needle tip secured with the needle in FIG. 7A, and the needle tip removed from the needle and held by the tool in FIG. 7B;
- (10) FIGS. 8A and 8B are cross sectional side views of another embodiment of a needle tip installation and removal tool, and needle, according to the present disclosure, with the needle tip secured with the needle in FIG. 8A, and the needle tip removed from the needle and held by the tool in FIG. 8B;
- (11) FIG. 9 is a front view of a needle tip installation and removal tool kit according to the present disclosure; and
- (12) FIG. 10 is a perspective view of a medical device comprising an access port with two needles, with each needle having a removable tip.

DETAILED DESCRIPTION

- (13) It is to be understood that this disclosure is not limited in its application to the details of construction and the arrangement of components set forth in the following description or illustrated in the drawings. The invention(s) herein are capable of other embodiments and of being practiced, or of being carried out, in various ways. Also, it should be understood that the phraseology and

terminology used herein is for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless limited otherwise, the terms “connected,” “coupled,” and “mounted,” and variations thereof which may be used herein are used broadly and encompass direct and indirect connections, couplings, and mountings. In addition, the terms “connected” and “coupled” and variations thereof are not restricted to physical or mechanical connections or couplings. Also, spatial references such as “above,” “below,” “top,” “bottom,” “horizontal,” “vertical,” “right,” “left,” and the like are meant to be understood in relation to the orientation of the device and components thereof as illustrated in the figure being described, and are interchangeable upon spatial reorientation of the device.

(14) By way of general overview, the present disclosure provides medical devices, systems and methods, and more particularly provides indwelling access devices, systems and methods which may comprise an implantable, subcutaneous, indwelling access port, which may be coupled with an implantable, subcutaneous, indwelling catheter within a body, particularly a human body. The access port may include an access port body containing an exposable/concealable internal needle that may extend/retract from the access port body to expose/conceal the internal needle.

(15) The present disclosure also provides medical grade needles with removable tips, as well as methods of needle tip replacement, particularly installation and removal, and devices therefor. Needle tip replacement (installation and removal) devices may include needle tip removal tools and kits, including hand held and manipulated tools that facilitate the installation and removal of needle tips, particularly on access port needles, such as for vascular access ports.

(16) Referring now to the figures, and in particular FIGS. 1A and 1B, there is shown a first embodiment of a medical device **10** according to the present disclosure implanted in a body of a host **50**, such as a patient or other subject, who may be undergoing medical treatment or diagnosis. Medical device **10** may comprise an implantable (subcutaneous), indwelling access port **20** and an implantable (subcutaneous), indwelling catheter **30** coupled to the access port **20**. As shown, the medical device **10** is implanted beneath the surface **62** of tissue **60**, such as cutaneous (skin) tissue, and the indwelling catheter **30** may extend from the access port **20** through the wall **64** and into a lumen **66** of a blood vessel **68** in the tissue **60** of the host **50**. In such manner, lumen **145** of catheter body **144** and lumen **66** of blood vessel **68** are in fluid communication. In such application, access port **20** may be understood to comprise a vascular access port.

(17) As shown in FIG. 1A, the access port **20** includes a pointed, closed tip, hollow needle **110** contained and housed within access port body **100**. As shown in FIG. 1A, the needle **110** is in a retracted position relative to the access port body **100**, and may include a pointed, removable, atraumatic, dialator tip **111** removably coupled to a distal end **114** (relative to the access port **20**) of the needle shaft **112**, which closes the distal end of shaft lumen **113**. As shown in FIG. 1B, needle **110** is in an extended position relative to the access port body **100**, and the removable pointed tip **111** has been removed to expose and provide access to lumen **113**.

(18) The needles **110** contemplated herein may particularly have gauge sizes in a range of 12 gauge to 22 gauge (e.g. 12, 13, 14, 15, 16, 17, 18, 19, 20, 21 or 22 gauge). However the needles may also have other sizes, small or larger. The needles **110** may exhibit an outer diameter in the range of 0.1 mm to 4.6 mm, including all values and increments therein. In addition, the needles **110** may exhibit an inner diameter in the range of 0.08 mm to 4.0 mm, including all values and increments therein. Furthermore, the needles **110** may exhibit a nominal wall thickness in the range of 0.002 mm to 0.4 mm, including all values and increments therein. The needles **110** may be formed of stainless steel or other materials.

(19) As explained in greater detail herein, the removable tip **111** may be removably secured to the needle shaft **112**, such as by threaded engagement of opposing complementary threads on each component. For example, referring briefly to FIG. 5B, the distal end portion of the needle shaft **112** may include external threads **154** which mate and engage within internal threads **156** of needle

shaft receptacle **158** of needle tip **111**. As such, when needle tip **111** is rotated in a first direction (e.g. clockwise or counter-clockwise) relative to the needle shaft **112**, the needle tip **111** may be installed (mechanically engage) on needle shaft **112**. Conversely, when needle tip **111** is rotated in a second direction opposite the first direction (e.g. counter-clockwise if the first direction is clockwise and clockwise if the first direction is counter-clockwise), the needle tip **111** may be removed (mechanically disengage) from needle shaft **112**. Stated another way, the complementary threads **154**, **156** may be either right-hand threads or left-hand-threads.

(20) Referring back to FIGS. **1A** and **1B**, a proximal end **116** of the needle shaft **112** may be secured within the access port body **100** against inadvertent removal from the access port **20**. The needle **110**, including either needle tip **111** and/or needle shaft **112**, may be made of a magnetic (ferromagnetic) material, or may include a ferromagnetic material at the distal end **114**. Examples of ferromagnetic material include iron, nickel and/or cobalt.

(21) The access port body **100** may include an external surface **104** and an internal fluid flow passage **101** which may allow fluid to flow through the access port **20** in either direction as described herein. Access port body **100** may include a fluid containment chamber **102**, which may optionally provide a reservoir for fluid to be stored within the access port **20**. The access port **20** may also include first and second bores **106**, **107** on different sides of the chamber **102** which provide tubular passages **108**, **109**, respectively, connecting or providing fluid communication between the chamber **102** and the external surface **104** of the access port **20**. Needle **110** may be positioned within bore **107**/passage **109** and may extend from and/or retract into the access port body **100** relative to a outer self-closing seal **132**, such as a “self healing” silicone septum described below. Thus, within access port body **100**, the internal fluid flow passage **101** which allows fluid to pass through access port **30** may be formed by chamber **102**, tubular passage **108** and lumen **113** of needle **110**.

(22) To extend the needle **110** as shown in FIG. **1B**, an actuator **40** comprising a magnet **M**, such as an electro-magnet, may be positioned closely over (adjacent within about 10 mm, and more particularly within about 5 mm) or on the needle **110**, or it may be positioned closely over or on a device to which the needle **110** may attach. When an electric current of a first polarity is provided to magnet **M**, magnet **M** may emit an electro-magnetic field arranged with a first polarity which attracts needle **110**. The needle **110**, being attracted (pulled) towards the magnet **M** by the electro-magnetic force of the electro-magnetic field, may be exposed by extending outwards from the access port body **100** of the access port **20** towards the magnet **M** and out of the host **50** by piercing through the skin surface **62** from within tissue **60**.

(23) The needle tip **111** of the needle **110** is designed to operate as a dilator during and after the distal (terminal) pointed end of the needle tip **111** (which may be referred to as a pencil tip) has penetrated through tissue **60**. With the configuration as shown, the needle tip **111** of the needle dilates the tissue **60** rather than cutting through the tissue **60** to minimize injury.

(24) Alternatively, when it becomes desirable to retract the needle **110** back into the access port body **100** of the access port **20**, after the needle tip **111** has been placed on needle shaft **112**, an electric current of a second polarity opposite the first polarity (i.e. reverse polarity) is provided to magnet **M**. Magnet **M** may then emit an electro-magnetic field arranged with a second polarity which repels the needle **110** from the magnet **M**, in which case the needle **110** will be pushed away from the magnet **M** by the force of the electromagnetic field and retract inwards relative to access port body **100** and the host **50** to be concealed. As the needle **110** travels inwards in access port body **100**, needle **110** correspondingly retracts and withdraws into cutaneous (skin) tissue **20**.

(25) In certain embodiments, after the needle **110** is extended, rotation of the needle **110** may lock the needle **110** in place against retraction. For example, the proximal end **116** of the needle **110** may include a projection **120** that may engage or otherwise cooperate with a needle lock mechanism **122**, such as by rotating onto a catch or into a channel, provided in the chamber **102** at a predetermined location.

(26) In addition, in certain embodiments the magnet M may be positioned on or within the device to which the needle **110** may be affixed to administer a given fluid (liquid) composition. For example, the magnet M may be positioned proximal to the lip of a vial, near the vial stopper, or in the tip of a catheter into which distal end **114** of the needle **110** may be asserted.

(27) In certain embodiments, a needle extension biasing mechanism **124**, such as a spring, may be positioned between the proximal end **116** of the needle **110** and a chamber wall **126** to retain the needle **110** in the retracted position. As may be appreciated, the force $F(s)$ exerted by the spring **124** on the needle **110** towards the retracted position may be less than the force $F(m)$ exerted by the magnet M, or stated another way, the force $F(m)$ is greater than the force $F(s)$.

(28) In certain embodiments, the needle **110** may include a flare **128** at the proximal end **116** which the spring **124** biases against when compressed and needle **110** is in the extended position. In addition, in certain embodiments, a bumper seal **130** may be provided to receive the proximal end **116** of the needle **110** in the retracted position to cushion retraction of the needle **110**. The bumper seal **130** may also function to close the proximal end **116** of the needle lumen **113** to prevent back flow of fluid through the needle **110** when the needle **110** is retracted, particularly in the event the needle **110** does not include pointed tip **111**. The bumper seal **130** may be formed into the chamber **126** or may be adhered onto the chamber walls.

(29) In certain embodiments, a self-closing seal **132**, such as a “self-healing” silicone septum, may be provided at the outer end **140** of tubular passage **109**. This seal **132** may be provided alone, or in addition to the bumper seal **130**, provided in the chamber **102**. The seal **132** may include a perforation **134** to allow the needle **110** to more easily pass through upon application of the magnetic force by the magnet M of the actuator **40**.

(30) In addition, in certain embodiments, an additional seal **136** may be provided at the inner end **138** of tubular passage **109** to prevent backflow of the fluid in the chamber **102** into the tubular passage **109**. It may be appreciated that further seals may be provided between the inner end **138** of the tubular passage **109** and the outer end **140** of the tubular passage **109**. In other embodiments, an expandable and/or collapsible sleeve **142** may be provided over the spring **124** and/or needle **110** preventing mingling of the fluids in the port with the spring surfaces or the exterior surfaces of the needle **110**. The sleeve **142** may be accordion like or in the shape of a bellows.

(31) Indwelling catheter **30**, and more particularly catheter body **144** may be removably attached to the access port **20** by a connector **146**, or permanently attached to the access port **20** through chemical or mechanical means, including an adhesive, ultrasonic welding, press-fits, etc. The catheter body **144** may be relatively flexible and formed of a composition such as silicone, polyurethane, or other thermoplastic elastomers. In addition, in some embodiments, a metering device **150** may be provided between chamber **102** of the access port **20** and the catheter **30**. The metering device **150** may include a valve and allow for control of the flow rate of fluid through the access port **20**.

(32) Referring now to FIGS. 2-4, there is shown needle **110** with removable needle tip **111** secured to the needle shaft **112**. A hand-held, hand-manipulated, needle tip installation and removal tool **300** for manipulating needle tip **111**, particularly installing and removing a needle tip **111** on needle shaft **112** is shown in FIG. 2. Tool **300** comprises an elongated tool body **301** having a proximal end **302** and a distal end **304** (relative to the user of tool **300**). Tool body **301** may include an elongated handle **306**, a shaft **308** extending from the handle **306**, and a needle tip holder **310** located at a distal end of the shaft **308**.

(33) The handle **306** may have an ergonomically designed shape which may include gripping features **314** that facilitate the gripping and manipulation of the tool **300** by hand, particularly by the palm and fingers. For example, the handle **306** may include a plurality of equally spaced, elongated, alternating ridges **316** and grooves **318** that may extend along a portion of the length of the handle **306** or substantially the whole length of the handle **306**. The handle **306** may generally have a length in a range of 3 to 7 inches, and be the same diameter as the shaft **308** or larger. The

shaft **308** may be cylindrical and also have a length of 3 to 7 inches.

(34) The needle tip holder **310** may comprise a tubular (e.g. polygonal, circular) needle tip receptacle **320**, located at distal end **304** of the needle tip installation and removal tool **300**, configured to receive and accommodate the needle tip **111** which resides at the distal end **114** of a needle shaft **130**. The needle tip **111** may have a mating portion **160** configured to mate and interact with a mating portion **322** of the needle tip receptacle **320** of the tool **300** for easy installation onto or removal from the needle shaft **112** (which may also be referred to as a cannula) of the access port **20**.

(35) More particularly, the needle tip receptacle **320**, particularly the internal sidewall of the mating portion **322**, may have a peripheral shape configured to mate and interact with the peripheral shape of the external sidewall of mating portion **160** of the needle tip ill.

(36) Even more particularly, the internal sidewall of the mating portion **322** of the needle tip receptacle **320** and the external sidewall of mating portion **160** of the needle tip **111** may mate and interact such that, when tool **300** is rotated about its longitudinal rotational axis LRA in a first direction (e.g. clockwise) the needle tip **111** may be installed on needle shaft **112**, particularly by the internal threads **156** of the needle tip **111** engaging with the external threads **154** of the needle shaft **112**. Conversely, when tool **300** is rotated about its longitudinal rotational axis LRA in a second direction opposite the first direction (e.g. counter-clockwise) the needle tip **111** may be removed from needle shaft **112**, particularly by the internal threads **156** of the needle tip **111** disengaging from the external threads **154** of the needle shaft **112**.

(37) As shown in FIGS. **4**, **5A** and **5B**, the internal sidewall of the mating portion **322** of the needle tip receptacle **320** and the external sidewall of mating portion **160** of the needle tip **111** may each comprise a plurality of planar sections **324** and **162**, respectively. The plurality of the planar sections **162** of the mating portion **160** of the needle tip **111** may be arranged around a periphery of mating portion **160** to form a polygonal shape. A shown, the plurality of the planar sections **162** of the mating portion **160** of the needle tip **111** may more particularly form a six-sided (male) polygon, i.e. a hexagon. Similarly, the plurality of the planar sections **324** of the mating portion **322** of the needle tip receptacle **320** may more particularly form a six-sided (female) polygon, i.e. hexagon.

(38) In order to hold needle tip **111** in needle tip holder **310**, particularly when the threads **154** of the needle shaft **112** and threads **156** of needle shaft receptacle **158** of needle tip **111** are disengaged during installation or removal of needle tip **111**, at least a portion of the needle tip holder **310**, and more particularly the needle tip receptacle **320**, may be formed by magnet **330**, such as an electro-magnet or permanent magnet.

(39) In the foregoing manner, when the needle tip **111** is in needle tip holder **310**, the needle tip **111** may be held in the needle tip holder **310**, and more particularly the needle tip receptacle **320**, by the magnetic force of the magnet **330**. In the case of an electro-magnet, the needle tip **111** may be removed from the needle tip receptacle **320** by shutting off the electromagnet or, in the case of a permanent magnet, by exposing the needle tip **111** to an opposite stronger magnetic force. Alternatively, the needle tip **111** may be removed from the needle tip receptacle **320** simply by pulling the needle tip **111** out of needle tip receptacle **320** by hand, against the bias force of the permanent magnet **330**.

(40) Referring now to FIGS. **6A** and **6B**, in another embodiment of the present disclosure, in order to hold needle tip **111** in needle tip holder **310**, particularly when the threads **154** of the needle shaft **112** and threads **156** of needle shaft receptacle **158** of needle tip **111** are disengaged during installation or removal of needle tip **111**, the tool **300** may include a vacuum (pressure below atmospheric pressure) passage **340** which may extend through handle **306**, through shaft **308** and to needle holder **310**, particularly to needle tip receptacle **320**. To provide vacuum to needle holder **310**, and more particularly the needle tip receptacle **320**, vacuum passage **340** may be in fluid communication with a vacuum device **350**, such as a vacuum generating device or vacuum source

device, which may be part of tool **300**, such as located in handle **306**. The needle tip **111** may then be removed from the needle tip receptacle **320** by turning off or reducing the vacuum. Alternatively, the needle tip **111** may be removed from the needle tip receptacle **320** simply by pulling the needle tip **111** out of needle tip receptacle **320** by hand, against the bias force of the vacuum.

(41) Referring now to FIGS. 7A and 7B, in another embodiment of the present disclosure, in order to removably hold needle tip **111** in needle tip holder **310**, particularly when the threads **154** of the needle shaft **112** and threads **156** of needle shaft receptacle **158** of needle tip **111** are disengaged during installation or removal of needle tip **111**, at least a portion of the needle tip holder **310**, and more particularly the mating portion **322** of the needle tip receptacle **320**, may form a friction fit (which also may be known as an interference fit or pressure grip fit) connection with the needle tip **111**. A friction fit connection may be understood herein as a connection formed between two components which solely relies upon friction to inhibit separation of the components, for example by one of the components being pressed into the other component such that at least one of the components is compressed (deformed) against one another.

(42) In order to provide a friction fit connection, a portion of the needle tip receptacle **320**, and more particularly the mating portion **322**, may be formed with a smaller dimension than the mating portion **160** of the needle tip **111** to form the friction fit connection. As shown, at least a portion of the mating portion **322** of needle tip receptacle **320** may be formed by an elastically deformable member **360**, such as rubber, which may deform (as to enlarge) when the needle tip **111** is located therein to hold the needle tip **111**.

(43) Referring now to FIGS. 8A and 8B, in another embodiment of the present disclosure, in order to hold needle tip **111** in needle tip holder **310**, particularly when the threads **154** of the needle shaft **112** and threads **156** of needle shaft receptacle **158** of needle tip **111** are disengaged during installation or removal of needle tip **111**, at least a portion of the needle tip receptacle **310**, and more particularly the mating portion **322** of the needle tip receptacle **320**, may form a positive mechanical connection with the needle tip **111**. A positive mechanical engagement connection may be understood herein as a connection formed between the components which does not rely solely on friction to inhibit separation of the components and which includes a mechanical interlock to inhibit separation of the components (e.g. overlapping surfaces).

(44) In order to provide a positive mechanical engagement connection, the mating portion **160** of the needle tip **111** may include external threads **370** which mate and engage within internal threads **380** of the mating portion **322** of the needle tip receptacle **320** of the needle tip holder **310**. As such, when tool **300** and the needle tip receptacle **320** are rotated in a first direction (e.g. clockwise or counter-clockwise) relative to the needle tip **111**, the needle tip receptacle **320** may mechanically engage with the needle tip **111**. Conversely, when tool **300** and the needle tip receptacle **320** are rotated in a second direction opposite the first direction (e.g. counter-clockwise if the first direction is clockwise and clockwise if the first direction is counter-clockwise), the needle tip receptacle **320** may mechanically disengage from the needle tip **111**. Stated another way, the threads **370**, **380** may be either right-hand threads or left-hand-threads.

(45) If, in the event, the removable tip **111** is positively mechanically engageable to the needle shaft **112** by threaded engagement of threads **156** and threads **154**, respectively, and the needle tip receptacle **320** is positively mechanically engageable to the needle tip **111** by threaded engagement of threads **380** and threads **370**, respectively, the pairs of threads **156**, **154** and **380**, **370** should have opposite drive hands. Thus, if threads **156**, **154** are right-hand threads, then threads **380**, **370** should be left-hand threads, and vice-versa.

(46) In another embodiment, similar to the embodiment of FIGS. 7A and 7B, at least a portion of the needle tip receptacle **320** of the embodiment of FIGS. 8A and 8B may be formed of an elastically deformable material, such as rubber, which may deform (as to enlarge) when the needle tip **111** is located therein to hold the needle tip **111**. In such instance, the alternating circumferential

ridges and grooves provided by the threads **380** may provide positive mechanical engagement by ratchet or snap fit connection, which may be in addition to or as an alternative to the positively mechanical engagement being provided by threads **380** which are rotationally engageable and disengageable. In such an instance, the ridges and grooves provided by the threads **380** of the needle tip receptacle **320** and threads **370** of the needle tip **111** may now operate linearly as rows of engageable/disengageable teeth for a linear sliding ratchet. More particularly, as the needle tip **111** is pushed into the needle tip receptacle **320**, the elastic threads **380** of the mating portion **322** of the needle tip receptacle **320** may deform as deformable teeth, such as with a sliding snap fit, allowing the threads **370** of the needle tip **111** to enter the needle tip receptacle **320** past the threads **380** of the needle tip receptacle **320** without requiring rotational action. In such instance, it may be understood that the positive mechanical engagement is formed by a portion of the needle tip **111** which overlaps a portion of the needle tip receptacle **320** in a plane transverse to a longitudinal axis of the tool **300**.

(47) For the foregoing positive mechanical engagement embodiment, as well as the friction fit connection, tool **300** may include a passage similar to passage **340** which extends through handle **302**, through shaft **308** and to needle holder **310**. A push rod may then be inserted into the passage to apply pressure to the needle tip **111** to push the needle tip **111** from the needle tip receptacle **320** and dislodge the tip from the tool **300**. Alternatively, a button may be disposed at the proximal end of the handle **306** which, when depressed, may activate a member extending through the passage to dislodge the needle tip **111** from the tool **300**.

(48) It should be understood that any of the foregoing structures and methods of retaining needle tip **111** in needle tip holder **310**, particularly when the threads **154** of the needle shaft **112** and threads **156** of needle shaft receptacle **158** of needle tip **111** are disengaged during installation or removal of needle tip **111**, may be combined into a single tool **300** as suitable. For example, the mating portion **160** of the needle tip **111** and the mating portion **322** of the needle tip receptacle **320** of the needle tip holder **310** may be temporarily retained to one another by releasable positive mechanical engagement/connection or friction fit engagement/connection. Such may be supplemented with magnetic and/or vacuum assist.

(49) It should also be understood that the mating portion **160** of the needle tip **111** and the mating portion **322** of the needle tip receptacle **320** of the needle tip holder **310** may be circular, particularly threaded, as shown in FIGS. **8A** and **8B**, or non-circular, as shown in FIGS. **2** to **7B**, in which case the mating portions may have a polygonal shape (e.g. triangular, quadrilateral (e.g. square), pentagonal hexagonal, heptagonal, octagonal, etc.)

(50) It should also be understood that the mating portion **160** of the needle tip **111** and the mating portion **322** of the needle tip receptacle **320** of the needle tip holder **310** may have one or more circumferential ridges and grooves, particularly provided by threads, as shown in FIGS. **8A** and **8B**, or have planar sections, as shown in FIGS. **2** to **7B**, in which case the mating portions may have a polygonal shape.

(51) Referring to FIG. **9**, disposable or reusable kits **400** may be provided specifically configured for single use per operation of the access port **20**, particularly with sterile tools **300** and sterile tips **111** residing on a sterile tray **404** sealed with heat sealed sterile packaging **402**. Generally, kit **400** may have two or four sterile tools **300**. For an access port **20** with only one needle **110**, the kit **400** may include a tray **404** including two sterile tools **300** and optionally one sterile tip **111** (one tool **300a** to remove the tip **111** and the other tool **300b** to install a new tip **111**). For an access port **20** with two needles **110**, such as an access port **20** for hemodialysis (see FIG. **10**), the kit **400** may include a tray **400** including four sterile tools **300** and optionally two sterile tips **111** (two tools **300a** to remove the tips **111** and the two tools **300b** to install a new tips **111**).

(52) The kit **400** may also contain the replaceable sterile tips **111**, or these could be provided in their own separate packaging. The tools **300a** for removing the tips **111** may be the same or different from the tools **300b** for applying the new tips **111**. In some embodiments, the tools may be

color coded. The tray **404** may contain molded recesses **406** having a shape which contains the tools **300**, and a further recess **408** to facilitate the grasping of the tools **300** out of the tray **404**. In some embodiments, the needle tips **111** may be preloaded onto the tools **300** so as to be ready to use.

(53) The kits **400** need not contain the tools **300a** for the removal of the needle tips **111** as sterility is not essential at that stage of the procedure. Hence, the sterile kit **400** may include at least one tool **300** with the needle tip **100** pre-fitted in the needle tip receptacle **320** of the tool **300b** to be applied to the needle shaft **112** of the access port **20** without further manipulation than opening a kit packaging, grasping the tool **300b** and applying the tip **111** onto the needle shaft **112**. In certain embodiments, the sterile kit **400** includes two tools **300b** with each tool **300b** having fitted thereon a needle tip **111**. In certain embodiments, the sterile kit **400** includes the access port **20**.

(54) The tools **300** may be made of material for single use or capable of withstanding autoclaving for further use. The tools **300** may be made of moldable polymer, such as polycarbonate or polyphethyleneoxide, or an appropriate metal, such as stainless steel and/or titanium, or other materials depending on the sterilization requirements.

(55) In certain embodiments, the tool **300** may be configured to incorporate the functions of both removal of a needle tip **111** and installation of a new tip **111**. Such design may include, for example, a tool **300** that has two needle tip receptacles **320**, one on each opposite end, one for removal of needle tips **111** and the other for installation of new tips **111**.

(56) In operation, once the access port **20** is in the configuration with the needle **110** protruding from the port **20** and through the skin, a packaging seal of the kit **400** is broken and a first tool **300** is used to remove the first needle tip **111**. In the case of an access port with two needles **110**, then a second tool **300**, or the same tool **300**, is used to remove the second needle tip **111**.

(57) The exposed needle shaft(s) **112** is/are then connected to the treatment (therapy or diagnostic) device. Once the treatment is completed, a third tool **300**, or the same tool **300**, is used to install a new sterile tip **111** onto the first needle shaft **112**, and, in the case of an access port **20** with two needles **110**, then a fourth tool **300**, or the same tool **300**, is used to install a new sterile tip **111** onto the second needle shaft **112**. Hence, to guarantee the best sterile environment during these procedures of accessing the access port **20**, it is preferable to use a separate tool **300** for each needle **110** and each operation.

(58) While a preferred embodiment of the present invention(s) has been described, it should be understood that various changes, adaptations and modifications can be made therein without departing from the spirit of the invention(s) and the scope of the appended claims. The scope of the invention(s) should, therefore, be determined not with reference to the above description, but instead should be determined with reference to the appended claims along with their full scope of equivalents. Furthermore, it should be understood that the appended claims do not necessarily comprise the broadest scope of the invention(s) which the applicant is entitled to claim, or the only manner(s) in which the invention(s) may be claimed, or that all recited features are necessary.

Claims

1. A medical device comprising: at least one needle tip having a pointed closed end configured to penetrate tissue, the needle tip configured to be installed on a hollow needle shaft having a fluid flow passage within the needle shaft, and the needle tip configured to be removed from the needle shaft; the needle tip configured to close the fluid flow passage within the needle shaft when the needle tip is installed on the needle shaft, the at least one needle tip is securable to the needle shaft by a positive mechanical engagement connection; and at least one tool configured to install the at least one needle tip on the needle shaft, the tool comprising a needle tip holder, the needle tip holder comprising a needle tip receptacle configured to receive the needle tip, including the pointed closed end, therein, and configured to hold the needle tip during an installation of the needle tip on the

needle shaft.

2. The medical device of claim 1 wherein: the at least one needle tip and the at least one tool are components of a sterilized medical device kit.
3. The medical device of claim 1 wherein: the positive mechanical engagement connection comprises a threaded connection.
4. The medical device of claim 3 wherein: the at least one needle tip comprises a needle shaft receptacle to receive a portion of the needle shaft; the needle shaft receptacle includes needle shaft receptacle threads; the needle shaft includes needle shaft threads; and the at least one needle tip is securable to the needle shaft by threaded engagement of the needle shaft receptacle threads of the needle tip with the needle shaft threads of the needle shaft.
5. The medical device of claim 1 wherein: the needle tip receptacle of the needle tip holder comprises a mating portion configured to mate with a mating portion of the needle tip.
6. The medical device of claim 5 wherein: the mating portion of the needle tip receptacle is configured to mate with the mating portion of the needle tip such that, when the tool is rotated about a longitudinal axis of rotation, both the needle tip receptacle and the needle tip rotate about the longitudinal axis of rotation.
7. The medical device of claim 5 wherein: the mating portion of the needle tip receptacle has a polygonal shape; and the mating portion of the needle tip has a polygonal shape which fits inside the polygonal shape of the needle tip receptacle.
8. The medical device of claim 1 wherein: the needle tip receptacle of the needle tip holder is configured to hold the needle tip during the installation of the needle tip on the needle shaft.
9. The medical device of claim 8 wherein: the needle tip receptacle is configured to hold the needle tip during the installation of the needle tip on the needle shaft with a friction fit connection formed with the needle tip.
10. The medical device of claim 9 wherein: the needle tip receptacle includes an elastically deformable member; and the friction fit connection is formed by elastic deformation of the elastically deformable member when the needle tip is in the needle tip receptacle.
11. The medical device of claim 8 wherein: the needle tip receptacle is configured to hold the needle tip during the installation of the needle tip on the needle shaft with a magnetic force applied to the needle tip.
12. The medical device of claim 11 wherein: the needle tip receptacle includes a magnet; and the needle tip comprises a magnetic material.
13. The medical device of claim 8 wherein: the needle tip receptacle is configured to hold the needle tip during the installation of the needle tip on the needle shaft with a vacuum force applied to the needle tip.
14. The medical device of claim 13 wherein: the needle tip receptacle is in fluid communication with a vacuum passage through which a vacuum is drawn with a vacuum generating device.
15. The medical device of claim 8 wherein: the needle tip receptacle is configured to hold the needle tip during the installation of the needle tip on the needle shaft with a positive mechanical engagement connection formed with the needle tip.
16. The medical device of claim 15 wherein: the at least one needle tip includes needle tip threads; the needle tip receptacle includes needle tip receptacle threads; and the positive mechanical engagement is formed by threaded engagement of the needle tip threads with the needle tip receptacle threads.
17. The medical device of claim 15 wherein: the at least one needle tip includes at least one needle tip ridge and/or at least one needle tip groove; the needle tip receptacle includes at least one needle tip receptacle ridge and/or at least one needle tip receptacle groove; and the positive mechanical engagement is formed by positive mechanical engagement of the at least one needle tip ridge with the at least one needle tip receptacle groove and/or engagement of the at least one needle tip groove with the at least one needle tip receptacle ridge.

18. The medical device of claim 15 wherein: the positive mechanical engagement is formed by a portion of the at least one needle tip which overlaps a portion of the needle tip receptacle in a plane transverse to a longitudinal axis of the tool.

19. The medical device of claim 1 wherein: the needle tip is configured to close the fluid flow passage within the needle shaft when the needle tip is installed on the needle shaft to ingress of fluid into the fluid flow passage.

20. The medical device of claim 1 wherein: the fluid flow passage within the needle shaft is provided by a lumen of the needle shaft.

21. The medical device of claim 20 wherein: the needle tip is configured to close the lumen of the needle shaft when the needle tip is installed on the needle shaft to ingress of fluid into the lumen.
